# Pace to pass

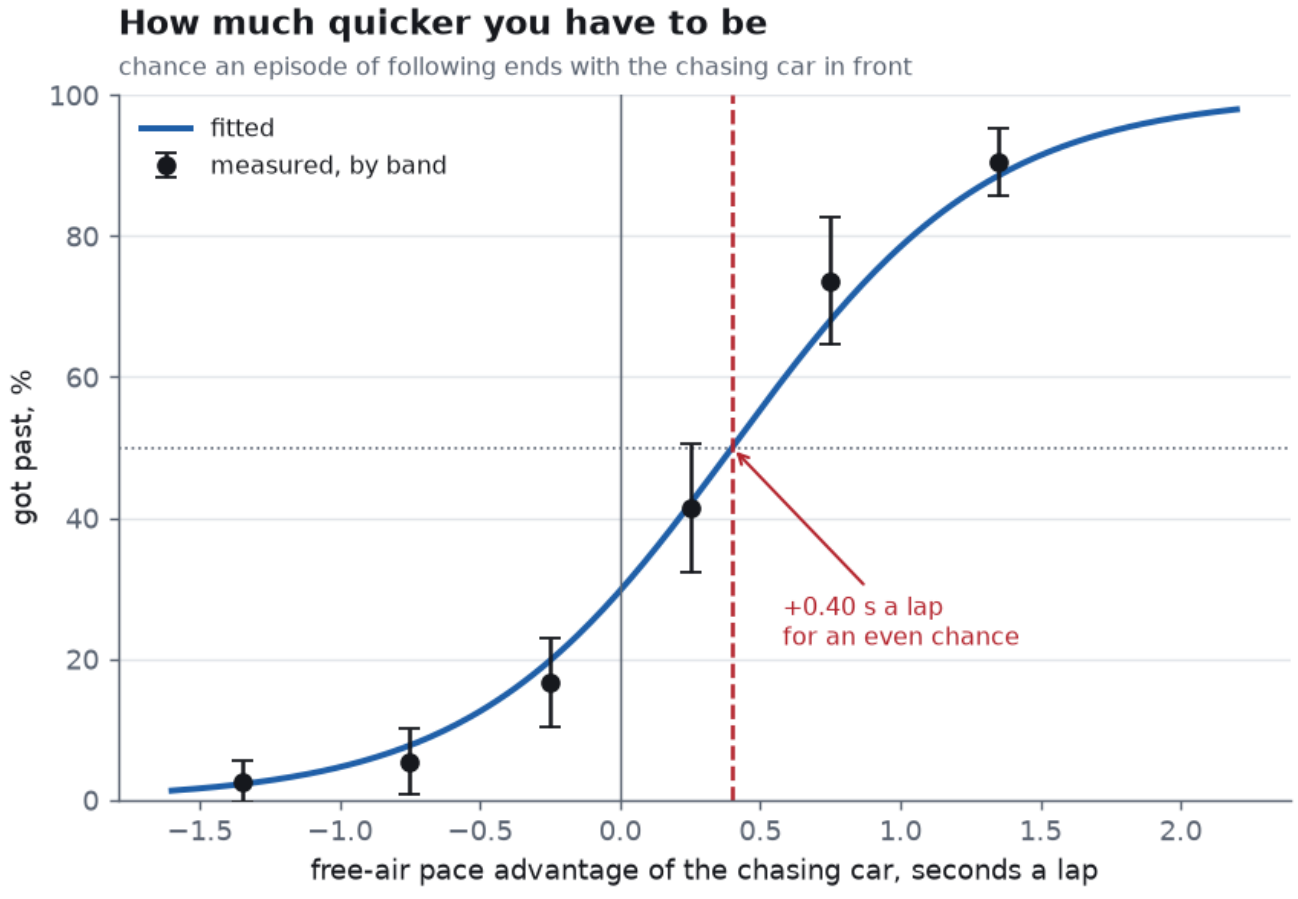
How much quicker a car has to be to get past the one in front, measured from 2023 lap times alone — and whether the circuit makes any difference.

## It takes +0.40 seconds a lap to get past

At the mean opening gap of 1.24 s, over 689 episodes. Being held up costs +0.25 s a lap, and the circuit makes no measurable difference (p 0.47).

| +0.40 s | 689 | +0.25 s | p 0.47 |
|---|---|---|---|
| pace needed to get past | episodes measured | cost of being held up | circuit differences |

The race order is in the lap times. Summing a car's lap times gives its elapsed race time, so sorting cars by that total at lap n puts them in the order they crossed the line. Against the position the feed reports, this agrees on 23,956 of 23,956 laps — 100.00%. Nothing here needs a positional feed, a gap feed or telemetry.



# Which pairs are measured

| outcome | episodes |
|---|---|
| pairs within reach | 10,515 |
| dropped: a pit lap | 1,619 |
| kept: never got past | 1,574 |
| kept: ended in a pass | 869 |
| dropped: the race ended first | 92 |
| dropped: a car stopped running | 34 |

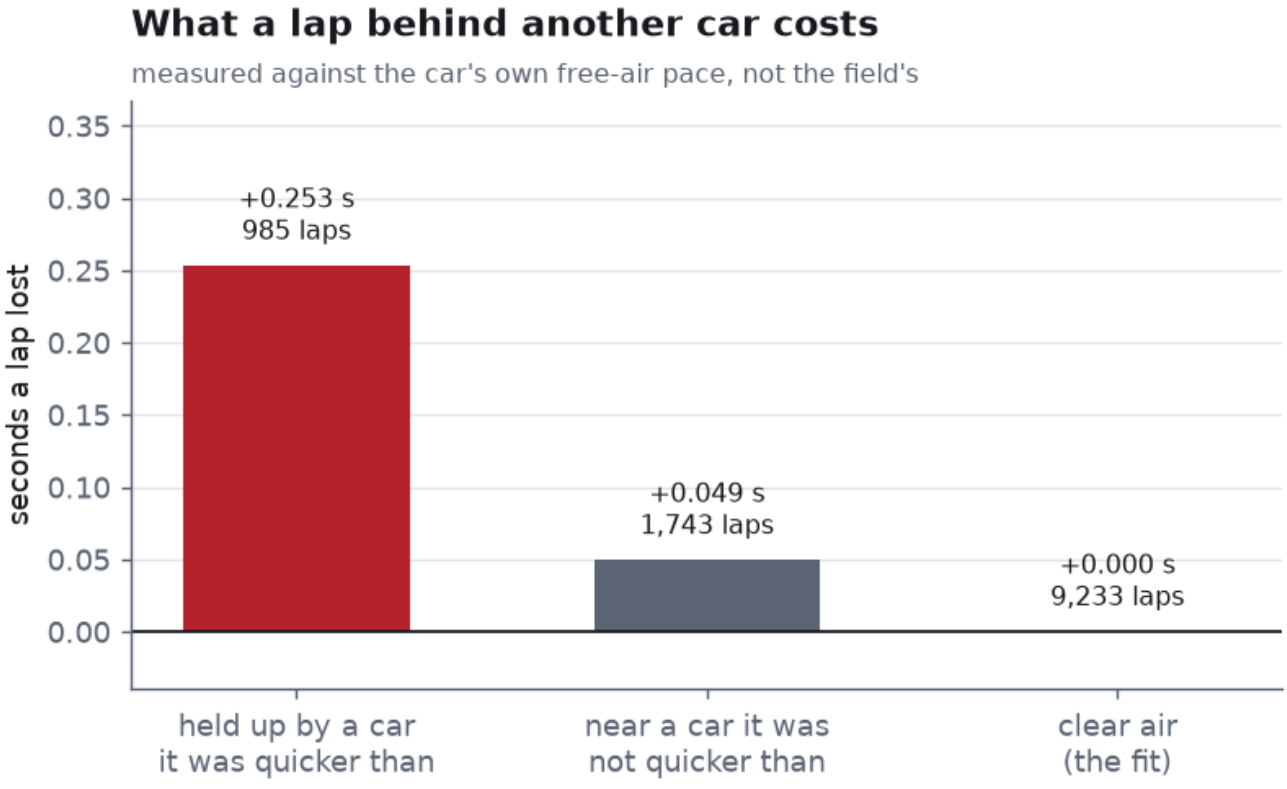
# How quick each car was

The dose cannot be read off the laps two cars spend together, because those laps are the outcome: a car stuck behind another is slowed by being stuck behind it. So pace is measured apart and tested together — each car-race gets a line fitted to the laps it ran in clear air, and the advantage is the difference of two lines, each read at that car's own tyre age. The fitted field spans 2.57 s a lap at the median race, which is what a Formula 1 grid covers.

| outcome | laps or car-races |
|---|---|
| clean laps | 18,769 |
| held out: in traffic | 8,193 |
| fitted car-races | 295 |
| refused: age and lap not separable | 57 |
| refused: too few clear laps | 25 |

## What being held up costs

The second bar is the control. Being close to another car costs nothing in itself — +0.049 s a lap for cars that were not quicker anyway. The cost appears only where the car behind had the pace to go faster and could not: +0.253 s a lap. Pooling the two, which is the obvious thing to do, dilutes it to almost nothing.



# How much quicker you have to be

At the mean opening gap of 1.24 s, the fitted curve crosses even odds at +0.40 seconds a lap. The intercept is +0.013: two cars of identical pace running nose to tail are a coin toss, which is what it ought to be and was not imposed.

| free-air advantage | episodes | got past |
|---|---|---|
| -9.0 to -1.0 | 115 | 2.6% ± 2.9% |
| -1.0 to -0.5 | 92 | 5.4% ± 4.6% |
| -0.5 to +0.0 | 132 | 16.7% ± 6.4% |
| +0.0 to +0.5 | 111 | 41.4% ± 9.2% |
| +0.5 to +1.0 | 91 | 73.6% ± 9.1% |
| +1.0 to +9.0 | 148 | 90.5% ± 4.7% |

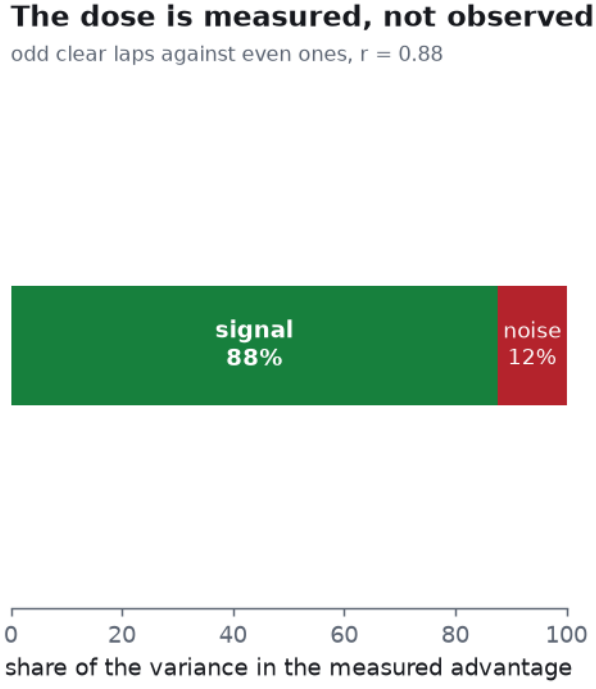
# The fitted curve

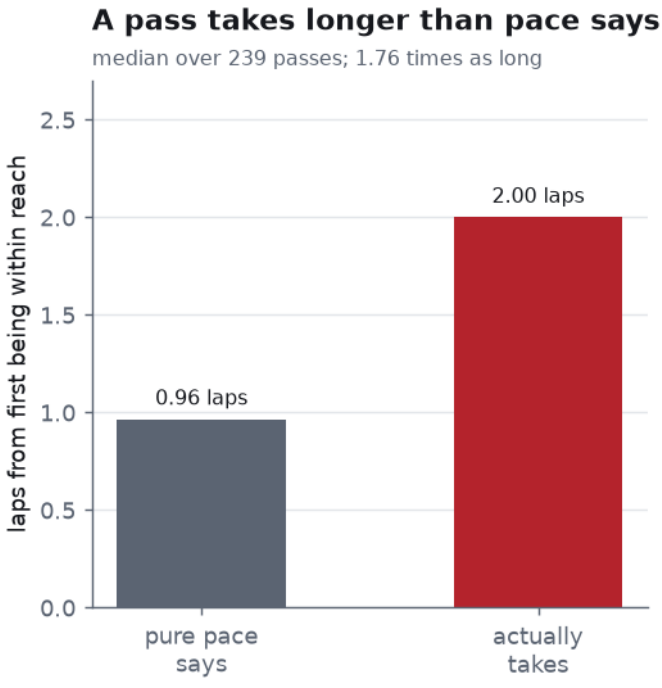
One logistic fit over every episode, with the opening gap as a control.

| term | estimate | 95% interval | p |
|---|---|---|---|
| base | +0.0125 | -0.5287 to +0.5538 | 0.9637 |
| per second per lap of advantage | +2.1552 | +1.7314 to +2.5790 | 0.0000 |
| per second of opening gap | -0.7067 | -1.1044 to -0.3091 | 0.0005 |

# The dose is measured, not observed

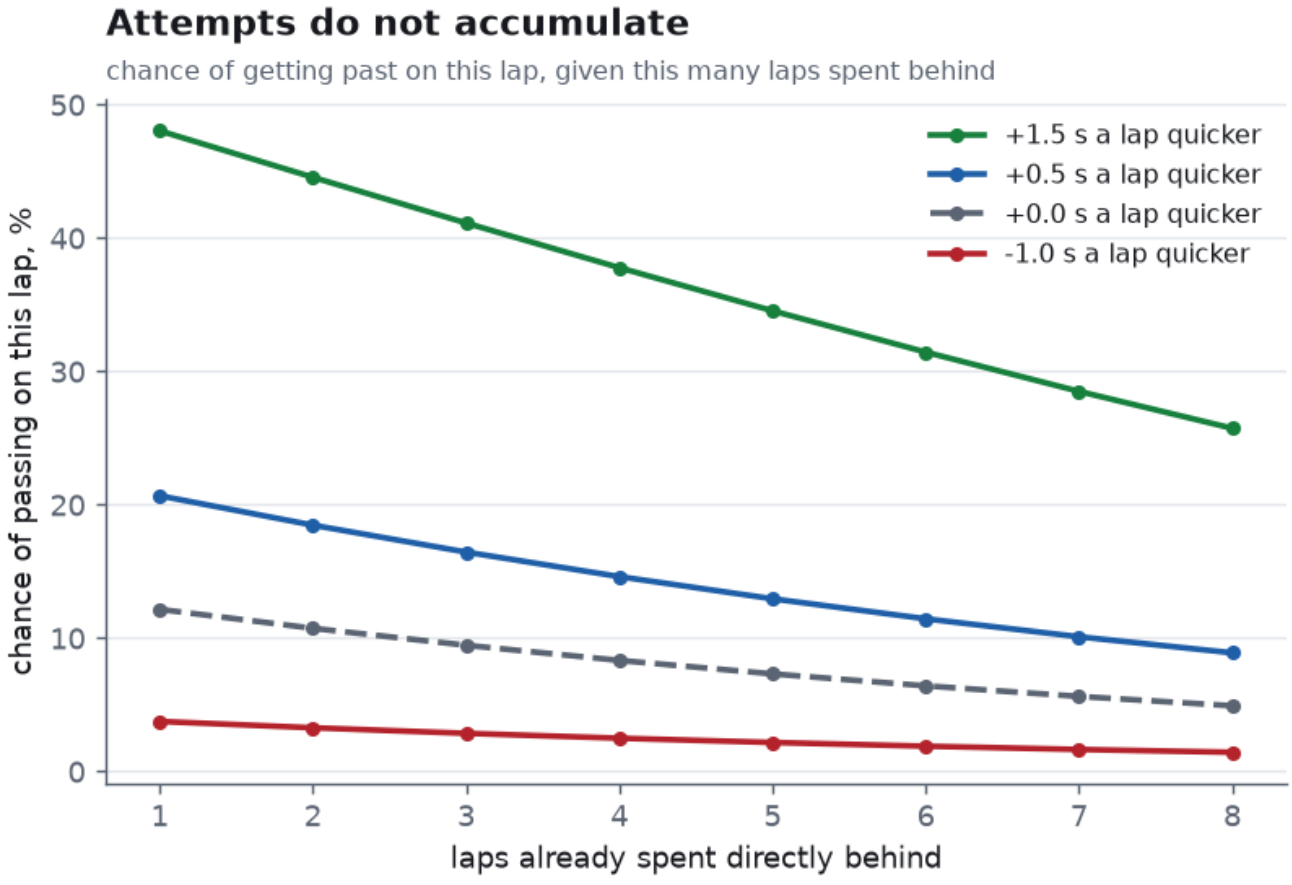
Fitting each car's line twice — once on its odd clear laps, once on its even ones — gives two independent measurements of the same quantity. They correlate 0.877, leaving noise of 0.376 s a lap against signal of 1.001. Reliability is 0.876, so the fitted slope is pulled toward zero by about that factor: the true slope is nearer 2.46 and the threshold nearer +0.35 s a lap.



## Attempts do not accumulate

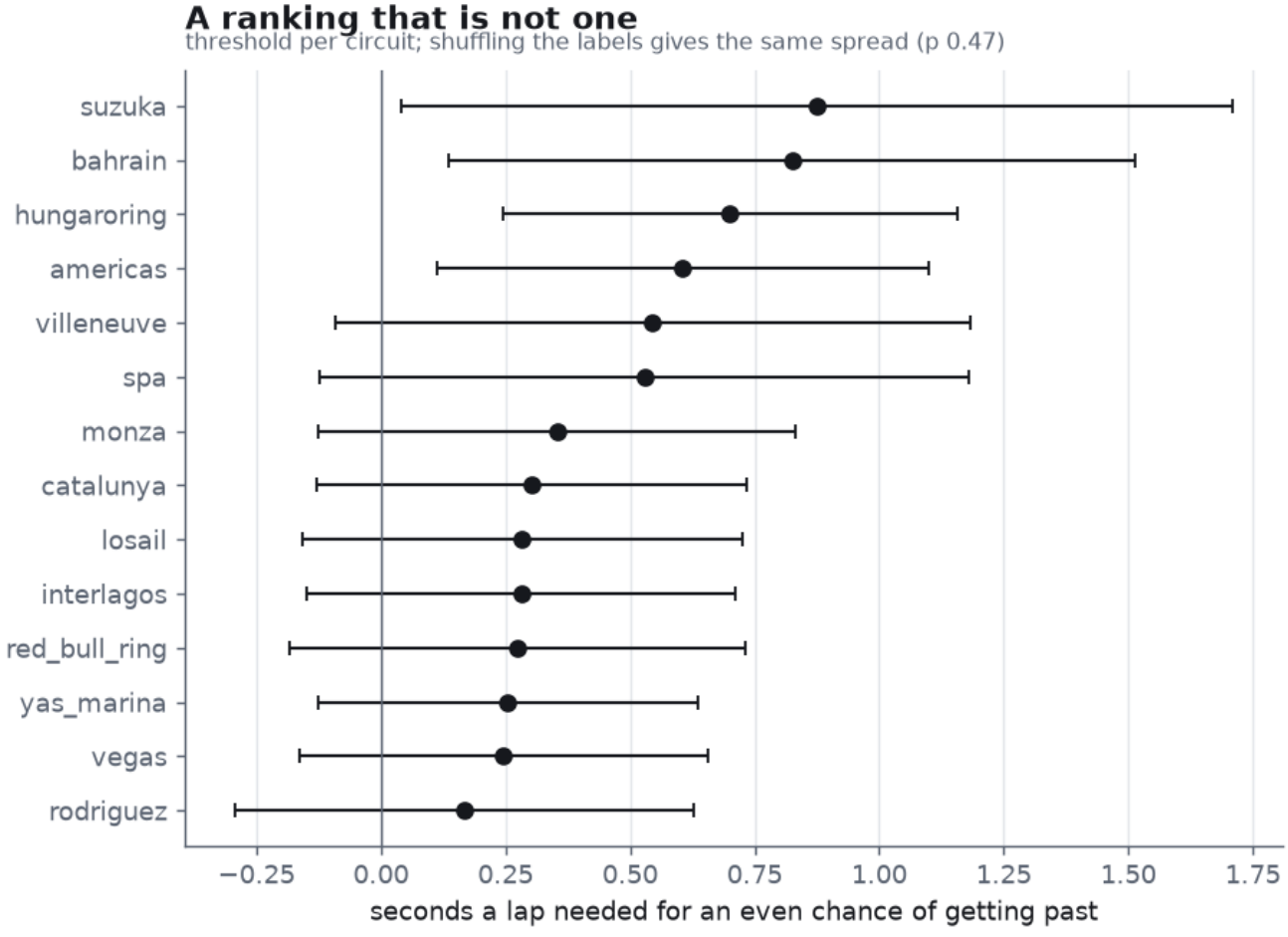
The time term is negative. A car that has not got past in the first couple of laps is less likely to manage it on the next one, not more. And a pass takes 1.76 times as long as pure pace says it should — 2.00 laps against 0.96 — which understates the resistance, because 33 more cars were still behind past the lap pace said they would be by and never got past at all.




| term | estimate | 95% interval | p |
|---|---|---|---|
| base | -0.8835 | -1.3272 to -0.4399 | 0.0001 |
| per second per lap of advantage | +1.2662 | +0.9654 to +1.5669 | 0.0000 |
| per lap of trying | -0.1402 | -0.2498 to -0.0306 | 0.0123 |
| per second of opening gap | -0.7691 | -1.0491 to -0.4890 | 0.0000 |

# Does the circuit matter? Not measurably

That column is not a ranking. Jointly, the 14 circuit terms give F 0.909, p 0.549. The fitted thresholds span 0.708 s a lap; shuffling the circuit labels across episodes — which destroys any real circuit effect and keeps everything else — gives a median span of 0.699, and beats the real spread 47% of the time. One season cannot resolve circuit differences, and the apparent ordering is what noise produces on its own.



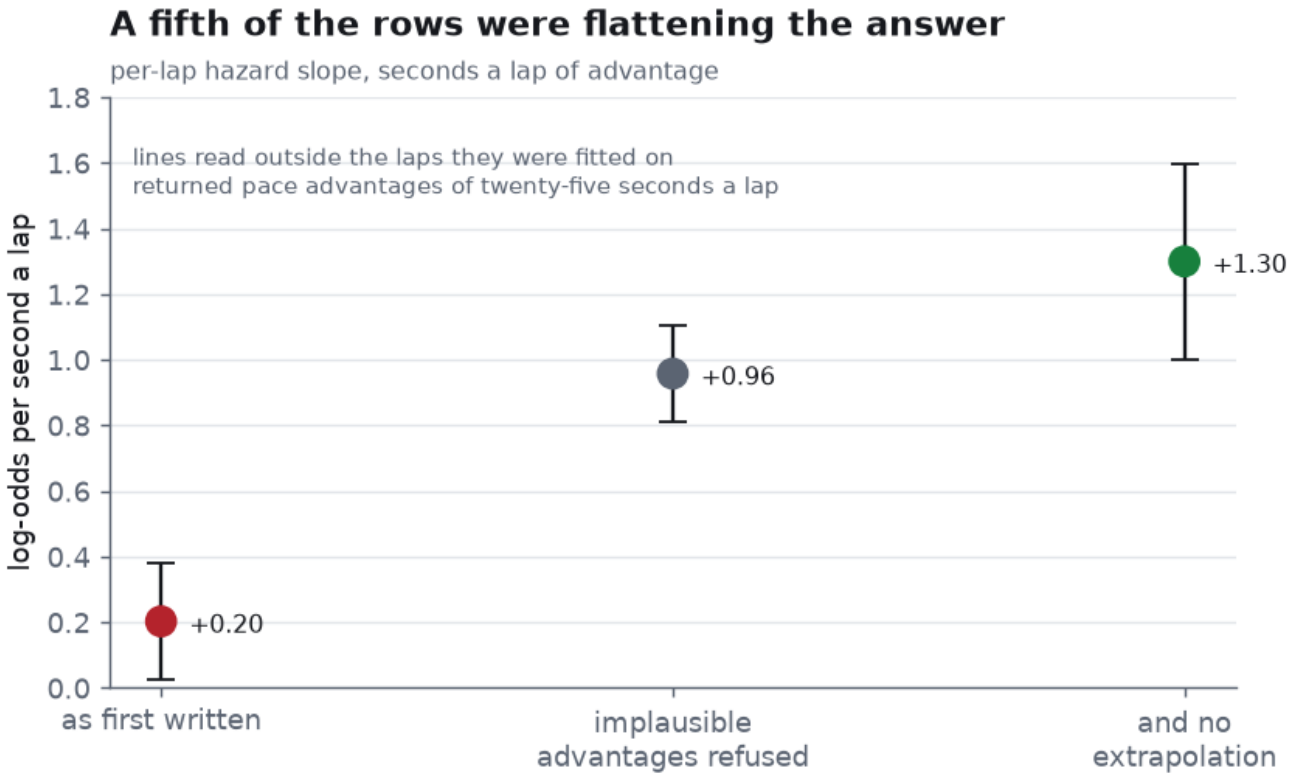

| circuit | threshold | 95% interval | episodes | got past |
|---|---|---|---|---|
| suzuka | +0.87 | +0.04 to +1.71 | 29 | 12 |
| bahrain | +0.83 | +0.14 to +1.51 | 31 | 9 |
| hungaroring | +0.70 | +0.25 to +1.16 | 43 | 16 |
| americas | +0.61 | +0.11 to +1.10 | 55 | 20 |
| villeneuve | +0.54 | -0.09 to +1.18 | 28 | 8 |
| spa | +0.53 | -0.12 to +1.18 | 39 | 16 |
| monza | +0.35 | -0.13 to +0.83 | 30 | 13 |
| catalunya | +0.30 | -0.13 to +0.73 | 52 | 27 |
| losail | +0.28 | -0.16 to +0.72 | 53 | 28 |

# Does the circuit matter? Not measurably (continued)

| circuit | threshold | 95% interval | episodes | got past |
|---|---|---|---|---|
| interlagos | +0.28 | -0.15 to +0.71 | 35 | 13 |
| red_bull_ring | +0.27 | -0.18 to +0.73 | 49 | 17 |
| yas_marina | +0.25 | -0.13 to +0.63 | 90 | 39 |
| vegas | +0.25 | -0.16 to +0.66 | 65 | 27 |
| rodriguez | +0.17 | -0.29 to +0.63 | 33 | 14 |

## What the guards were worth

A per-car line read outside the laps and tyre ages it was fitted on returned pace advantages of twenty-five seconds a lap. Two per cent of the rows were enough to flatten the central slope by a factor of five. Refusing implausible advantages recovers most of it; refusing extrapolation altogether recovers the rest, and the estimate then stops moving as the rule is tightened further.




Lap times and pit stops from the Jolpica mirror of the Ergast schema. The cleaning rules are carried over from the stint-degradation, race-strategy-mc and undercut-effect projects.